

DESIGN THINKING PROJECT

DOMAIN:-ZERO WASTE IN CAMPUS CAFETERIA

GROUP MEMBERS

- 1] Atharva Tupe CS7-02
- 2] Vishwajeet Jadhav CS7-12
- 3] Pratik Tapkir CS7-16
- 4] Aditya Gengaje CS7-20



Secondary Research

Zero-Waste in Campus Cafeteria

Campus cafeterias often generate significant food waste, contributing to environmental problems and financial losses. A zero-waste approach can transform food services, benefiting both the planet and the community.

Group Name: Zero-waste visionaries.

Members: Aditya Gengaje CS7-20

Vishvajeet Jadhav CS7-12

Pratik Tapkir CS7-16

Atharva Tupe CS7-02





This presentation outlines a secondary research-based approach to understanding and addressing zero waste campus by leveraging existing studies , reports and data, we can build empathy for users and identify

The Problem with Food Waste on Campus

Cafeteria food waste is a common problem on college campuses worldwide.

Students may overestimate their appetite or choose meals they don't finish, leading to food waste.

Poor food storage and inadequate portion sizes contribute to waste.

Environmental Impact of Food Waste

Greenhouse Gases

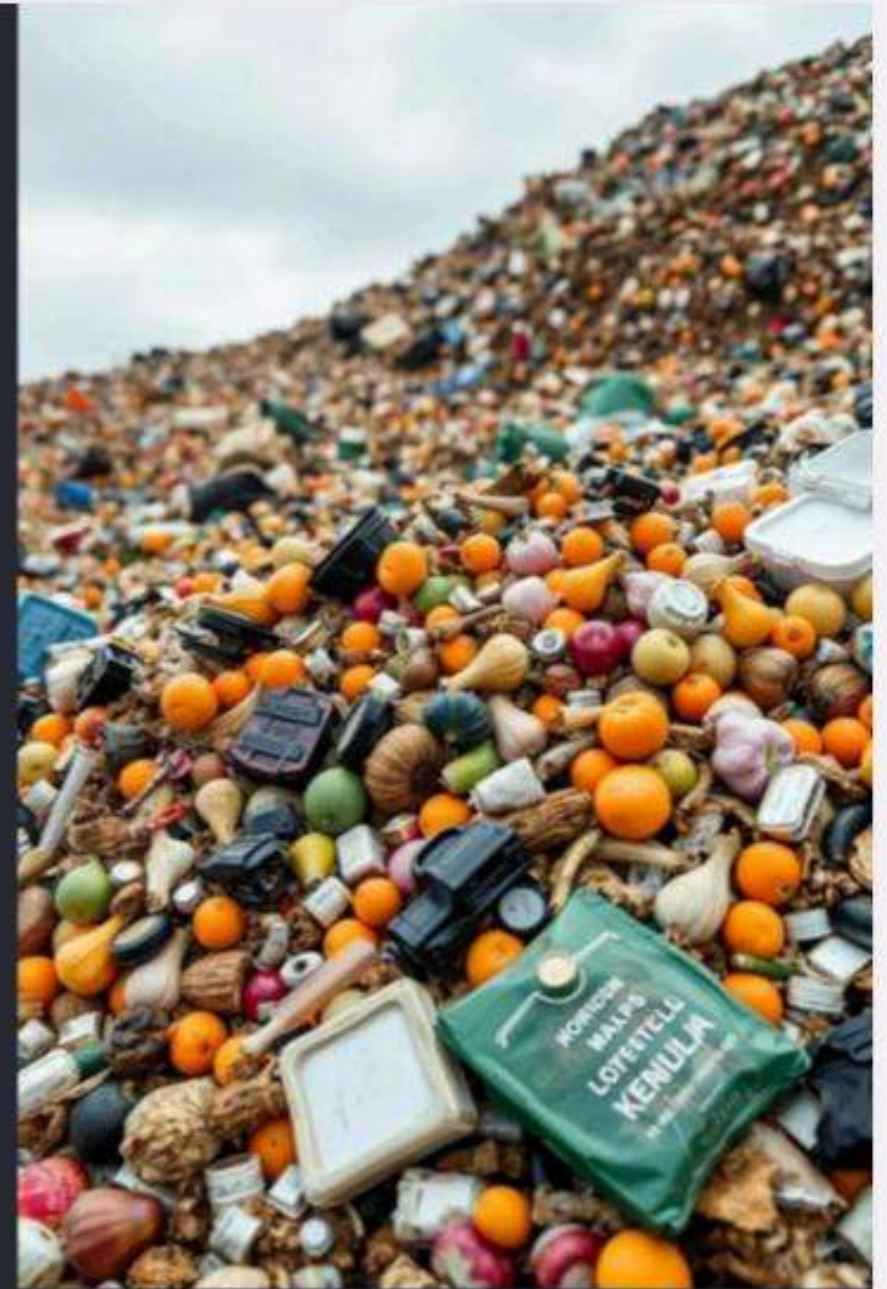
Landfills emit methane, a potent greenhouse gas contributing to climate change.

Water Waste

Growing food requires vast amounts of water, and wasted food represents wasted water resources.

Soil Degradation

Food waste in landfills depletes soil quality, impacting agricultural productivity.





Current Practices and Their Shortcomings

Landfill Disposal

Landfills are the primary disposal method, contributing to environmental problems.

Limited Recycling

Recycling programs often have limited scope and may not include food scraps.

Lack of Awareness

Students and staff may not be aware of the impact of food waste or have limited access to resources.

Introducing a Zero-Waste Cafeteria Model



Reduce

Implement strategies to minimize food waste from the start, such as portion control and food donation.



Recycle

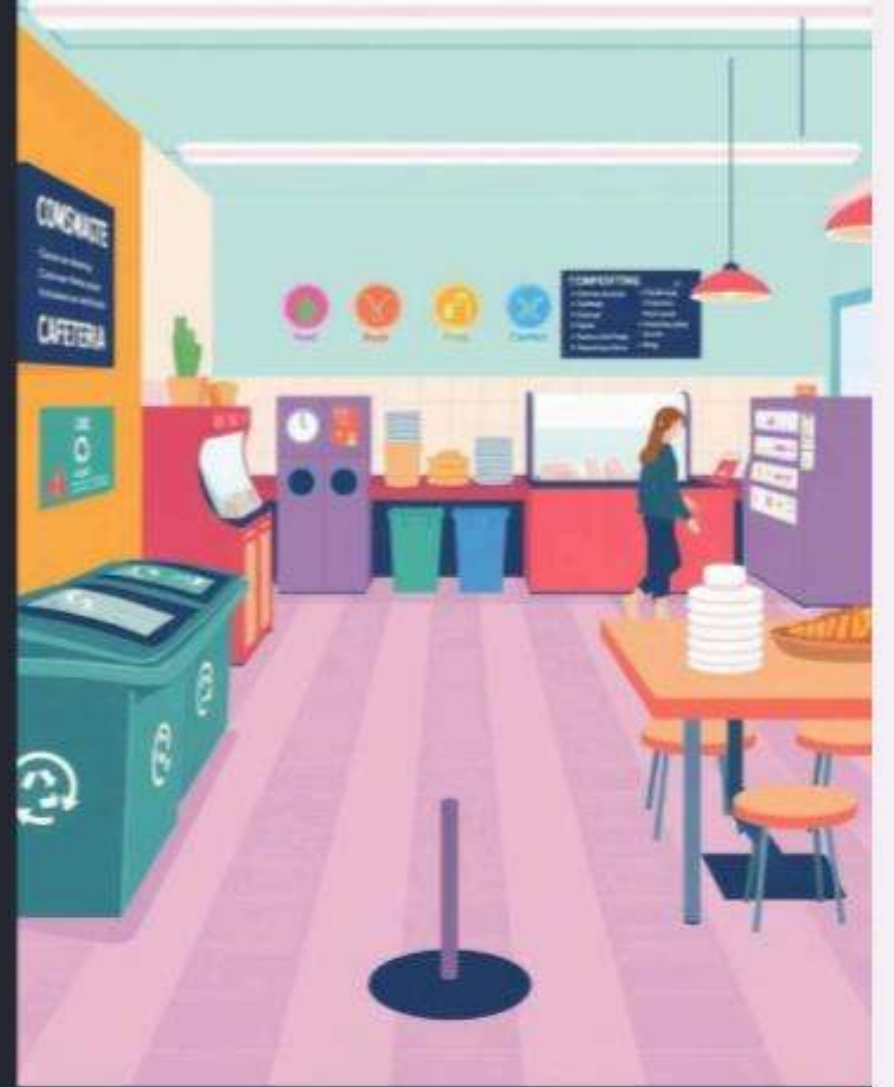
Establish robust composting and recycling systems for food scraps, paper, and other materials.



Reuse

Encourage the use of reusable containers and provide compostable food packaging.

ZERO-WASTE



Strategies for Reducing Food Waste

1

Portion Control

Encourage students to take smaller portions and offer smaller plate sizes.

2

Food Donation

Partner with local food banks or charities to donate surplus food.

3

Improved Food Storage

Implement proper storage practices to minimize food spoilage.





Composting and Recycling Initiatives

1

Composting

Set up designated composting bins for food scraps, coffee grounds, and other organic materials.

2

Recycling

Ensure appropriate recycling bins for paper, plastic, aluminum, and glass.

3

Divert Waste

Educate students on proper waste sorting to maximize composting and recycling.

Engaging Students and Staff



Measuring Success and Tracking Progress

1

Data Collection

Track food waste amounts, recycling rates, and composting efforts.

2

Performance Metrics

Measure the impact of the program on environmental and financial aspects.

3

Regular Reporting

Share progress updates with students, staff, and administrators.

Benefits of a Zero-Waste Cafeteria

1

Environmental Impact

Reduce greenhouse gas emissions and conserve resources.

2

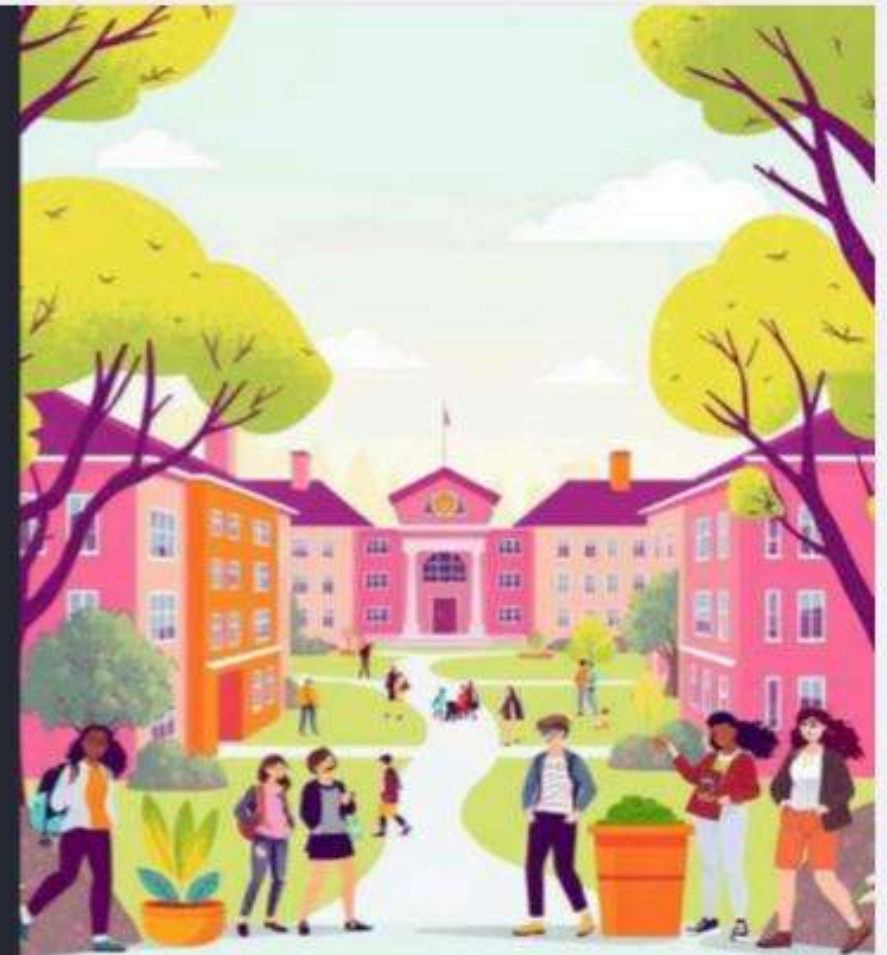
Financial Savings

Lower landfill fees and reduce food purchasing costs.

3

Community Engagement

Foster a sense of responsibility and pride in sustainability.



"Threats ccrose crapoar welcen rea d'ate, thare,
shot as you, zero-waiste to togl..

ZERO-WASTE



PRIMARY RESEARCH

- **Summary of project :-**

- **Purpose:**

- The goal of a zero waste campus cafeteria is to minimize waste and promote sustainability through thoughtful design and practices.

- **Key Components:**

- Reusable Items:** Durable plates, utensils, and cups.

- Composting Stations:** Areas for organic waste.

- Recycling Bins:** Clearly labeled for proper waste segregation.

- Sustainable Sourcing:** Locally and sustainably sourced food.

- Bulk Dispensing:** Minimizing packaging waste.

- Education and Awareness:** Programs to raise awareness.

- Food Recovery Programs:** Donating excess food.

- Water Stations:** Encouraging the use of reusable bottles.

- **Customer Revalidation:**

- Surveys:** Regular feedback collection.

- Observation:** Monitoring behavior.

- Engagement:** Hosting events and workshops.

- Testimonials:** Sharing customer stories.

- Continuous Improvement:** Acting on feedback.

A zero waste campus cafeteria aims to minimize waste and promote sustainability by incorporating reusable items, composting stations, and recycling bins. It prioritizes sustainable sourcing of food and bulk dispensing to reduce packaging waste.

Education and awareness programs play a crucial role in engaging the campus community and promoting zero waste practices. The cafeteria also includes food recovery programs to donate excess food and water refill stations to encourage the use of reusable bottles. Customer revalidation is achieved through regular surveys, observation, and engagement activities, ensuring continuous improvement based on feedback.

QUESTIONNAIRE :

How can we effectively reduce food waste in the cafeteria?

- a] Implement portion control,
- b] offer "take what you need" policies,
- c] create awareness campaigns.

What are the best practices for composting organic waste on campus?

- a] Set up compost bins,
- b] provide composting workshops,
- c] partner with local composting facilities.

How can we encourage students and staff to use reusable items instead of disposables?

- a] Offer discounts for bringing reusable items,
- b] provide reusable items at low cost,
- c] un awareness campaigns.

What strategies can be implemented to promote recycling and proper waste segregation?

- a] Clearly label recycling bins,
- b] run educational programs,
- c] provide incentives for proper recycling.

How can we involve the campus community in zero waste initiatives and education programs?

- a] Host events and workshops
- b] create a zero waste club,
- c] collaborate with student organizations.

What are some ways to source food and supplies sustainably for the cafeteria?

- a] Partner with local farmers
- b] prioritize organic products
- c] reduce packaging.

How can we manage and donate excess food to minimize waste?

- a] Partner with local shelters
- b] implement a food recovery program,
- c] use technology to track and manage surplus food.

What are the benefits of having water refill stations in the cafeteria?

- a] Reduce plastic waste
- b] provide convenient access to clean water
- c] encourage the use of reusable bottles.

How can we measure the success of our zero waste initiatives?

- a] Track waste reduction,
- b] conduct surveys
- c] monitor participation rates.

What incentives can be offered to encourage participation in zero waste practices?

- a] Provide rewards or discounts
- b] recognize participants publicly
- c] create a points system for sustainable actions.

How can we reduce the environmental impact of our food sourcing?

- a] Partner with local farmers
- b] prioritize organic products,
- c] reduce transport emissions.

What role can technology play in achieving zero waste goals?

- a] Use apps to track waste
- b] implement smart bins
- c] use data to optimize food orders.

How can we handle non-compostable waste effectively?

- a] Focus on recycling programs
- b] minimize non-compostable items,
- c] find eco-friendly alternatives.

What partnerships can enhance our zero waste efforts?

- a] Collaborate with local composting facilities
- b] work with environmental organizations
- c] partner with other schools

PHOTOS OF SURVEY





AEIOU Framework

- 1) A-Activity Sheet
- 2) E-Environment Sheet
- 3) I-Interaction Sheet
- 4) O-Object Sheet
- 5) U-User Sheet

AEIOU Framework	Group ID:	Sheet No: 1
Activity Sheet	Group Name: ZERO-WASTE VISIONARIES	
	Project Title: Zero-waste campus cafeteria.	
Things Happening Around Product Concept: 1.Fast food packages 2.Lack of dustbins 3.Uneaten food by people 4.Over production of food by café management 5.Food scraps and left over 6.single use of plastic plates 7.Production of non-recyclable material	Sketch/Photo- Summary of Activities 	
Effect of Activity on Product, Special Notes 1.Food packages thrown pollutes campus. 2.Waste of food causes financial cost. 3.Waste of resources like water, labour energy, food Ingredients. 4.Health and sanitation concerns. 5.Contributes to greenhouse gas emission. 6.Garbage waste impact climate change. 7.Requires more efforts to clean-up cafeteria.		

Environment Sheet

Group Name:

Project Title: Marker pen

Season, Time, Personal, Location For Product

Season effect:

Rainy: 1. Increased Indoor Dining: More indoor dining leads to higher use of disposable packaging
2. Higher Disposable Item Use: More paper towels and disposable rain gear are used.
3. Reduced Outdoor Activities: Outdoor seating and events are reduced, concentrating waste indoors.

Summer: 1. Increased Outdoor Dining: More disposable packaging used outdoors.
2. Higher Beverage Consumption: More disposable cups and bottles.
3. More Food Waste: Heat causes faster food spoilage.

Scenes /Photos of Environment Related to Concept



Winter: 1. Increased Indoor Dining: More people eat indoors, raising waste from packaging and disposables.

2. More Hot Beverages: Increased use of disposable cups, lids, and stirrers for hot drinks.

3. Higher Energy Consumption: More heating may increase energy waste.

Autumn: 1. Transition to Indoor Dining: As the weather cools, more students eat indoors, increasing disposable packaging waste.

2. More Leaf and Plant Debris: Fallen leaves and plant debris contribute to outdoor waste.

3. Increased Food Waste: Seasonal foods like pumpkins and squash may lead to more food scraps.

Spring: 1. More Outdoor Dining: As weather improves, more people dine outside, increasing disposable packaging waste.

2. Increased Food Waste: Seasonal food items may lead to more leftover food.

3. More Plastic Waste: More bottled beverages consumed during warmer weather



TIME EFFECT:

Day time: Peak dining leads to food overproduction.

Morning: More breakfast packaging waste.

Afternoon: Increased snack and beverage waste.

Evening: Larger portions result in food waste.

Night: night takeout increases packaging waste.

Late night: Waste management oversight reduces.

Early morning: Unchecked waste from the previous night.

Location effect: Waste increases in high-traffic or indoor areas due to higher disposable item usage.

Effect/Precaution For Product Based on Environment:

1. There should be proper management of food waste.

2. We should make limited food.

3. We should take required amount of food.

4. We must organize food donating campaign for needy poor peoples and beggars.

Interaction Sheet

Group Name:

Project Title:

Users Expectation About Product:

1. Convenient waste sorting
2. Sustainable food packaging
3. Affordable and accessible
4. Eco-friendly food options
5. Reusable containers to reduce disposables

Photos of Market Survey & Interaction:



Summary of Interaction:

1. User need cafeteria clean and in low cost and easy to maintenance so that anyone can use it properly and effectively.
2. User need easy to use waste sorting station for recycling, composting and trash, making waste disposal more convenient.
3. User prefer to see the cafeteria take initiatives to reduce food waste, energy consumption, and use of plastic.

Object Sheet

Group Name:

Project Title:

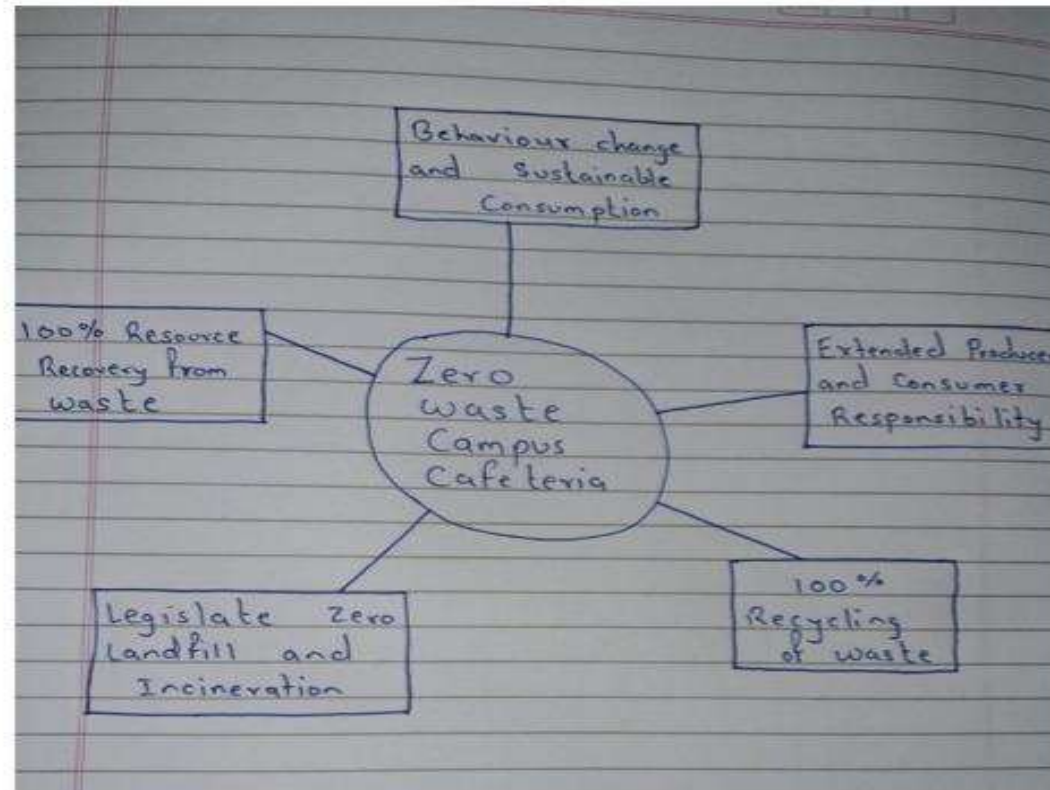
List of Components:

- 1.Compost bins.
- 2.Recycling stations
- 3.Reusable coffee mugs.
- 4.Digital menu
- 5.Smart waste bins
- 6.Portable utensils
- 7.Waste free events

List of Inventory with Photos :



Diagram Showing Interaction Between Objects:



User Sheet

Group Name:

Project Title:

Identified Potential Users of Product:

1. Students: Looking for affordable, sustainable, and convenient dining options.
2. Faculty and Staff: Regularly using the cafeteria for meals and seeking eco-friendly choices.
3. Environmentally Conscious Individuals: Those actively avoiding single-use plastics and waste.
4. Cafeteria Management: Staff responsible for implementing sustainable practices and ensuring efficient operations.

Existing User Photographs / Special Notes



5. Local Food Producers: Suppliers of sustainable and locally sourced ingredients.

Special Modification For Users/ Method of Use:

1. Reusable Container Program: Users bring their own or borrow reusable containers.
2. Seasonal Menu Adjustments: Seasonal, sustainable meals based on available ingredients.
3. Mobile App Integration: Pre-order meals and choose sustainable options via an app.
4. Composting Stations: Separate bins for composting food scraps.
5. Smart Waste Sorting: Automated bins that guide proper waste sorting.





Ideation Sheet

- **POV STATEMENT:-**

We believe a zero-waste cafeteria is essential to provide students and staff with a sustainable dining experience that reduces waste, lowers costs environmental responsibility.

- **SCAMPER:-**

- **S – Substitute**

Identify elements to replace with sustainable alternatives:

Single-use items → Reusables: Replace plastic plates, cups, and cutlery with reusable ceramic, glass, or stainless steel options.

Plastic packaging → Eco-friendly materials: Use compostable wraps (e.g., beeswax) or refill stations instead of pre-packaged items.

Paper towels → Cloth napkins: Washable cloth napkins for hand drying and cleaning.

- **C - Combine**

systems or processes to reduce waste:

Bulk Serving + Refill Stations: Combine self-serve dispensers for beverages, grains, or condiments to eliminate packaging waste.

Reusable Deposit System + Discounts: Combine reusable container systems with financial incentives to promote adoption.

Food Waste Collection + Composting: Integrate composting bins alongside food waste reduction programs.

- **A – Adapt**

ideas from other sustainable practices:

Bring Your Own (BYO) Programs: Encourage students and staff to bring their own reusable bottles, lunchboxes, and cutlery.

Circular Economy Models: Partner with local suppliers who use zero-waste or returnable packaging.

Zero-Waste Cafés: Learn from zero-waste cafes (like those using cup deposit schemes) to implement effective strategies.

- **M – Modify**

Make changes to improve efficiency and sustainability:

Portion Sizes: Offer flexible portion options to minimize food waste.

Menus: Modify the menu to include locally sourced, seasonal, and plant-based foods to reduce environmental impact.

Cooking Practices: Use leftovers creatively (e.g., turning vegetable scraps into broth).

- **P - Put to Another Use**

Repurpose or upcycle waste and resources:

Food Waste → Compost: Use cafeteria food scraps to create compost for campus gardens.

Leftovers → Donations: Donate untouched surplus food to local shelters or students in need.

Upcycled Materials: Repurpose old containers for organizing storage or growing plants.

- **E – Eliminate**

Remove unnecessary elements generating waste:

No Single-Use Plastics: Ban single-use plastic straws, utensils, and plates.

Avoid Over-Packaging: Eliminate pre-packaged foods by offering meals served directly on reusable trays or containers.

Remove Waste Bins (Strategically): Encourage recycling and composting by limiting general trash bins.

- **R - Rearrange / Reverse**

Reorganize systems to promote zero-waste behaviors:

Reverse Flow: Implement a return system for dishes, cutlery, and containers to ensure reuse.

Rearrange Serving Style: Switch to buffet-style or bulk dispensers to reduce packaging.

Design Traffic Flow: Place waste sorting stations (recycle, compost, landfill) at exits to encourage proper disposal.

PUGH MATRIX :

Criteria	Base line: Disposable items	Reusable Dishes	Composting	Food waste Tracking	Education on Campaign
Cost	0	1	0	0	0
Fast of Implem- entation	0	-1	0	-1	1
Environ- mental impact	0	1	1	1	1
student satisf- action	0	1	0	0	1
Scalabi- lity	0	0	1	1	1
Total	0	2	2	1	4



Ideation Canvas

Ideation Canvas		Group ID:	Sheet No: 7
Group Name: ZERO-WASTE VISIONARIES		Project Title: ZERO-WASTE CAMPUS CAFETERIA (CAFETERIA ELITE SPACE)	
 People 1) VISITOR'S 2) STUDENTS 3) VENDORS AND SUPPLIERS 4) WASTE MANAGEMENT TEAM 5) CAFETERIA STAFF			
 Activities 1) EATING AND SOCIALIZING 2) STUDY AND GROUP PROJECTS 3) RELAXING AND LEISURE 4) MEAL SERVICE AND DINING 5) ORDERING		 Situation/Context/Location (SITUATION) : 1) LONG WAIT TIMES DURING PEAK HOURS 2) DIFFICULTY IN SORTING WASTE PROPERLY 3) GROUPS STRUGGLING TO FIND SEATS TOGETHER (LOCATION) : 际 CAMPUS CAFETERIA (CONTEXT) : 1) CAMPUS EVENTS. 2) REGULAR COLLEGE HOURS. 3) SEASONAL EVENTS. 4) LUNCH BREAK.	



Props/Possible Solutions

- 1) **REUSABLE CONTAINER:** Introduce a program where students can borrow reusable containers for a small deposit, which they can return or cleaning and reuse.
- 2) **COMPOSTING STATIONS AND FOOD WASTE BINS :** Set up designated compost bins for food scraps and biodegradable items in the cafeteria, accompanied by clear signage on what can and cannot be composted.
- 3) **REAL-TIME INVENTORY AND DEMAND TRACKING:** Use a digital system to track food inventory and monitor real-time demand, reducing overproduction and minimizing food waste.
- 4) **EDUCATIONAL CAMPAIGNS ON ZERO-WASTE PRACTICES:** Conduct awareness campaigns, workshops, or signage throughout the cafeteria to educate students on zero-waste practices. Topics could include proper waste sorting, the impact of food waste, and tips on bringing reusable containers.
- 5) **FOOD WASTE TO FERTILIZER SYSTEM:** Set up an on-site composting system for all organic food waste. This can be a simple bin or a more advanced composting machine that turns food scraps

Made with Xodo PDF Reader and Editor

into nutrient-rich compost, which can be used for local garden projects or plants in and around the canteen.

BRAINSTROMING IDEAS

1. Compostable Utensils and Plates: Use biodegradable or compostable plates, cups, and cutlery made from natural materials like bamboo or cornstarch.
2. Reusable Dining Options: Introduce a reusable dish and cutlery program with a deposit system for students and staff to borrow and return.
3. Food Waste Composting Stations: Set up food compost bins for organic waste that can be used for campus gardens or donated to local farms.
4. Leftover Food Donation Programs: Partner with local charities to donate unconsumed food daily.

5. On-Campus Recycling Stations:Create clear and accessible recycling bins for separating waste into plastics, paper, and glass.

6. Water Refill Stations:Install water refill stations to reduce single-use plastic bottle waste.

7. Educational Campaigns:Run workshops, posters, and events to educate students about the importance of zero waste and how they can participate.

8. Eco-Packaging Initiatives:Partner with suppliers to eliminate unnecessary packaging and use environmentally friendly materials.

MIND MAPPING

❑ ZERO WASTE IN CAMPUS CAFETERIA

1. Reducing Food Waste :

- Portion size customization (half/full servings)
- Pre-order meal systems
- Food donation programs
- Composting leftover food

2. Sustainable Packaging :

- Reusable containers and utensils
- Biodegradable packaging materials
- Bring Your Own Container
- Eco-friendly packaging partnerships

3. Recycling and Composting :

- Segregated waste bins with clear signage
- Recycling partnerships with local agencies
- Awareness campaigns for proper waste segregation



Empathy Map

Empathy Mapping

Group ID:

Sheet No: 6

Group Name: ZERO-WASTE VISIONARIES.

Project Title : ZERO-WASTE CAMPUS CAFETERIA (CAFETERIA ELITE SPACE)

Hear

- 1) A COMPLETE SURVEY MADE US UNDERSTAND THE FOLLOWING THINGS: 1) APPROXIMATELY 100-150 PEOPLE'S FOOD GET LEFT OVER DUE TO OVERPRODUCTION OF FOOD.
- 2) APPROXIMATELY 10-15 PEOPLES WORK IN CAFETERIA(5-6 COOKS , 2 SUPERVISOR AND REMAINING WORKERS
- 3) THE PROBLEMS FACED IS ABOUT THE VENDORS MIGHT DELIVER FOOD IN SINGLE-USE , NON-RECYCLABLE PACKAGING ADDING TO WASTE
- 4) MAINTENANCE OF COMPOSTING UNITS, DISHWASHERS AND PURCHASING ECO-FRIENDLY MATERIAL CAN BE EXPENSIVE.

Users

- 1) VISITORS
- 2) STUDENTS
- 3) VENDORS AND SUPPLIERS
- 4) WASTE MANAGEMENT TEAM
- 5) CAFETERIA STAFF

Think & Do

(THINK) :

- 1) IDENTIFY PEAK TIME, POPULAR ITEMS AND LEFT OVER FOOD TO UNDERSTAND WASTE SOURCES.
- 2) DIGITAL SCREENS, APPS OR NOTIFICATIONS COULD INFORM STUDENTS ABOUT MENU OPTIONS TO REDUCE OVERPRODUCTION.
- 3) PLAN FOR COMPOSTING RECYCLING FOR LEFT OVER FOOD OR UNAVOIDABLE WASTE.

(Do) :

- 1) PLACE COMPOST , RECYCLING AND WASTE BINS AROUND THE CAFETERIA.
- 2) INSTALL SPEAKERS TO INSTRUCT STUDENTS TO DON'T WASTE FOOD AND TAKE REQUIRED AMOUNT OF FOOD IN PLATES AS YOUR NEED.
- 3) CREATE RULES SO THAT STUDENTS WILL GET TO KNOW NOT TO WASTE FOOD. OTHERWISE PENALTY WILL BE TAKEN.

See & Feel

(SEE) :

- 1) WATCH CAFETERIA TRAFFIC DURING DIFFERENT TIMES TO IDENTIFY PEAK HOURS.
- 2) OBSERVE SALES DATA TO ANALYZE MOST POPULAR ITEM AND FREQUENTLY WASTED ITEM
- 3) SEE THAT COOK MUST MAKE REQUIRED AMOUNT OF FOOD SO THE FOOD WILL NOT REMAIN STALE .

(FEEL) :

- 1) FRUSTRATION AND TENSION.
- 2) DISCOMFORT FROM OVERCROWDING.
- 3) SPEND TIME IN KITCHEN TO FEEL THE WORK LOAD OBSERVING HOW RUSH FOOD PREPARED MAY LEAD TO OVER PRODUCTION.





Product Development

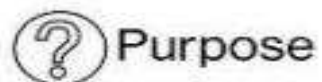
Product Development Canvas

Group ID:

Sheet No:8

Group Name: ZERO-WASTE VISIONARIES.

Project Title: ZERO-WASTE CAMPUS CAFETERIA.



Purpose

1. Reduce Waste.
2. Recycle and Compost.
3. Educate and Engage.
4. Promote Reusables.
5. Community Impact.



Components

1. Composting Stations
2. Recycling Bins
3. Sustainable Sourcing
4. Bulk Dispensing
5. Reusable Items



Product Features

1. Reusable Items
2. Composting
3. Recycling Bins
4. Sustainable Sourcing
5. Bulk Dispensing



People

1. Students
2. Staff
3. Visitors
4. Vendors and Suppliers
5. Waste management team



Product Experience

1. User-Friendly Design
2. Reliability
3. Support
5. Personalization
6. Feedback



Customer Revalidation

1. Phone Number
Verification: Add an extra layer of verification for users.
2. Manual Verification (If Needed): In case of suspicious activity or complaints, manually verify the user's identity.



Reject, Redesign, Retain

- Reject:** Single-use plastics, excessive packaging.
- Redesign:** Waste sorting, food distribution, educational initiatives.
- Retain:** Reusable items, sustainable sourcing, composting.



PROTOTYPE

Prototype :

In the prototyping phase of our design thinking project for a zero-waste campus cafeteria, we want to create practical solutions that reduce waste and encourage sustainability. Our first prototypes will include food and drink stations that use bulk dispensers, where students can refill reusable containers instead of using disposable packaging. This will help cut down on waste and promote eco-friendly habits.

We will also design reusable trays, plates, and cutlery, which will be collected, cleaned, and reused, so there's no need for disposable items. There will also be refillable water stations to reduce the use of plastic bottles.

Along with the physical prototypes, we will create signs and educational materials throughout the cafeteria to raise awareness about the importance of reducing waste. This will include posters, pamphlets, and digital displays that show the impact of food waste and plastic use. We'll also hold workshops to teach students and staff about sustainable practices.

During this process, we will involve students, faculty, and cafeteria staff in testing the designs and gathering feedback. This will help us improve the prototypes and make sure they work well for everyone. By working together and continuously improving, we aim to create a cafeteria that not only reduces waste but also inspires the campus community to adopt sustainable habits.



THANK YOU!!!